

Multigrid Self-Gravity in the AthenaK Shearing Box:

I. Mesh-Refinement Infrastructure (Phase 0), Shear-Periodic Boundary Conditions (Phase 1), and Gravity on Refined Meshes (Phases 2a–2b)

`athenak-multigrid tree, branch multigrid`

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Abstract

This document opens the *multigrid track* of the self-gravity program: the route to gas + dust self-gravity in AthenaK shearing boxes *with mesh refinement*. It records Phase 0 — the shearing-box/refinement infrastructure every later phase rests on: (i) an `orbital_advection` toggle with a fully consistent non-FARGO (full-velocity) mode; (ii) an enforced *interior- x_1* refinement policy replacing AthenaK’s blanket shearing-box+refinement prohibition; (iii) orbital advection (FARGO) on refined meshes under an *annular* refinement policy (refined regions span the full azimuthal extent); and (iv) *adaptive* refinement (Phase 0c): the shear-boundary communication state is rebuilt after every AMR update, a graded refinement cap keeps 2:1 balancing from ever refining the shear-periodic boundary columns, and an upstream derefinement-ordering bug that split annular rings is fixed. Validation: a refined-ring run reproduces a reference SMR run bit-for-bit in non-FARGO mode; with FARGO on the ring, mass is conserved exactly and a 20-orbit epicycle test tracks the analytic solution with an amplitude drift of only 0.008% (versus 0.064% on the uniform grid), with the epicyclic amplitude invariant flat to 10^{-4} . Under AMR, a prescribed moving ring that is created and destroyed every orbit for 20 orbits (2296 MeshBlocks created, 2240 destroyed) conserves mass exactly and matches the static-ring run an order of magnitude more closely than either matches the uniform grid; on the vortical-shwave problem the moving-ring solution converges to the static-ring one (4.7×10^{-2} apart at 64^2 , 1.8×10^{-4} at 128^2). Phase 1 (§6) then makes the multigrid Poisson solver itself shear-aware: shear-periodicity is imposed purely as a boundary *identification* — a y -remapped x_1 ghost fill applied at every level of the multigrid hierarchy through global boundary planes, with the 7-point operator untouched. Validated against the FFT oracle: solves at shear-periodic instants are bit-identical to plain-periodic solves; the analytic shearing-wave potential is recovered with errors *below* the FFT solver’s own (9.5×10^{-8} vs. 1.6×10^{-6} at 128^3); the swing-amplification history matches the FFT run to 2×10^{-7} ; and the V-cycle contraction factor — the one open risk named in Phase 0 — degrades only from 0.116 to 0.157 per cycle at a near-worst fractional shift, reaching the same 10^{-11} roundoff floor. Phase 2a (§7) puts the solver on *refined* shearing-box meshes with interior-ring static refinement. The never-exercised octet machinery inherited with Athena++ #776 is first anchored on the isolated binary-potential test of Tomida & Stone §4.2 (region-wise second order against the analytic two-sphere solution); opening the shear machinery to multilevel meshes whose x_1 boundary blocks remain at the root level then requires *no new shear-fill code* — only a relaxed guard. Validation: sheared-instant solves on the refined mesh are bit-identical to the periodic twin; the analytic shearing-wave error converges at second order in L_2 (first order at the max, which sits on the refinement interface exactly as in the periodic case); the swing test on a level-1 ring matches its uniform twin to the hydro truncation level — the gravity-off control deviates *more* than the gravity-on run — and multi-rank solves reproduce

serial to a float32 ulp. Phase 2b (§8) then admits refinement *on* the shear boundary (boundary annuli): the octets tiling the shear faces receive a y -remapped x_1 ghost fill — the Phase-1 plane algorithm at octet granularity, in host code with no MPI seam (octets are rank-replicated) — through scalar per-face twins of the validated remap kernels, and the block shear planes are sized at the boundary blocks’ level. Annuli solves at shear-periodic instants are bit-identical to their periodic twins at one and two refinement levels; the generic-phase L_2 equals the interior-ring value to four digits (the meshes differ by a half-box shift); contraction and floors match the periodic twins; and the annuli swing run matches its uniform twin to hydro truncation (again dominated by the gravity-off control) at machine-level phase lock. The companion document `fft_selfgravity_validation.pdf` covers the FFT solver used as the machine-precision oracle throughout this program; a Julia notebook (`multigrid_validation_tests.ipynb`) reproduces every test here.

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1 Plan and scope

1.1 Goal and roadmap

The scientific target is shearing-box simulations combining gas self-gravity, dust with mutual drag (streaming instability, including the background radial pressure gradient), and mesh refinement that zooms onto high dust-density regions in spiral arms launched by gravitational instability (cf. Baehr, Zhu & Yang 2022, ApJ 933, 100, who omitted the pressure gradient). Mesh refinement is the reason AthenaK was chosen, so the Poisson solver must ultimately live on refined meshes — which an FFT cannot. The adopted route therefore makes the *multigrid* solver shear-aware, with the validated FFT solver retained in three roles: machine-precision oracle for the multigrid shear boundary conditions, production solver for uniform-grid runs, and donor of exact open- z Dirichlet boundary values for stratified multigrid runs.

The phases of the multigrid track:

1. **Phase 0 (this document, done):** shearing box + refinement infrastructure. 0a: an `orbital_advection` on/off toggle with a consistent full-velocity (non-FARGO) mode, and the interior- x_1 refinement policy. 0b: FARGO on refined meshes under the annular refinement policy. 0c: adaptive refinement — dynamic creation and destruction of refined rings, without gravity, isolating the mesh machinery from the solver before Phase 2 couples them. Without 0b, refined shearing boxes pay a timestep penalty of $\sim (c_s + q\Omega L_x/2)/c_s \approx 10\text{--}30$ for gravito-turbulence boxes.
2. **Phase 1 (this document, §6, done):** shear-periodic boundary conditions in the multigrid driver (`MultigridDriver`), validated against the FFT oracle: exactness at shear-periodic instants, shearing-wave accuracy at generic shear phase, and the swing-amplification test with `solver=multigrid`. The V-cycle convergence rate with remapped boundaries — the one open risk of the program — is measured: the per-cycle contraction factor degrades only from 0.116 to 0.157.
3. **Phase 2:** multigrid gravity on refined shearing-box meshes (SMR first, then AMR), compared against high-resolution uniform FFT references.

4. **Phase 3–5:** merge with the `athenak-dust` particle-mesh dust module (dust density in the Poisson source, gravitational force on particles, pressure-gradient source term for the streaming instability), stratification with open- z boundaries, and the science runs.

1.2 Phase-0 scope and restrictions

What works after Phase 0, all enforced by fatal errors rather than conventions:

- Hydrodynamic shearing boxes with SMR, under the *uniform-boundary-level* rule: both shear-periodic x_1 boundary faces must share one refinement level. Refinement interior in x_1 always satisfies this (faces stay at root); alternatively *both* boundary faces may be refined uniformly (full x_2/x_3 extent, same level) — physically natural for a structure at the boundary, whose shear-periodic images live on both faces.
- Orbital advection on refined meshes, provided every refined region spans the full x_2 (azimuthal) extent — *annular rings* (interior, or boundary annuli as above). Without FARGO (`orbital_advection=false`) arbitrary refinement shapes are allowed subject only to the boundary-level rule.
- Adaptive mesh refinement (Phase 0c). The shear-boundary communication state (`ShearingBox::x1bndry_mbo` GID lists, MPI request arrays, CC/FC buffers) is rebuilt after every AMR update; AMR flag handling keeps rings intact (refine a ring if any member is flagged, derefine only if all are) and a *graded cap* refuses any refinement whose 2:1 balancing would propagate into the shear-periodic boundary columns (§5.8).

Not yet supported/exercised: MHD orbital advection with refinement (the face-centered EMF remap at level interfaces needs a $\nabla \cdot B$ analysis; fatal-guarded), multi-rank (MPI) exercise of the AMR rebuild path (the code is rank-aware but all Phase 0c runs are serial), and a production-size demonstration of the FARGO speedup itself. SMR/AMR runs must write `bin` outputs — the `vtk` writer does not support refined meshes.

2 Numerical method

2.1 Two frames for the shearing box

With background shear flow $\mathbf{v}_0 = -q\Omega x \hat{y}$, AthenaK can now evolve either of two variable sets, selected by `<shearing_box> orbital_advection`:

FARGO frame (`orbital_advection = true, default`). The stored velocity is the *fluctuation* $\mathbf{v}' = \mathbf{v} - \mathbf{v}_0$; the background advection is applied analytically once per step by the orbital-advection shift (Sec. 2.2). The momentum source terms are

$$\dot{M}_x = 2\Omega \rho v'_y, \quad \dot{M}_y = -(2 - q)\Omega \rho v'_x, \quad (1)$$

where the $(2 - q)$ coefficient is the sum of Coriolis (-2Ω) and the advection of background shear by the radial fluctuation ($+q\Omega$). The CFL condition sees only $|\mathbf{v}'|$.

Full-velocity frame (`orbital_advection = false`). The stored velocity includes the shear. The source terms are the full Coriolis force plus the tidal force, and (for an ideal EOS) the tidal work:

$$\dot{M}_x = 2\Omega \rho v_y + 2q\Omega^2 x \rho, \quad \dot{M}_y = -2\Omega \rho v_x, \quad \dot{E} = 2q\Omega^2 x \rho v_x \quad (2)$$

(the Coriolis force does no work). Because the wrapped data at the shear-periodic x_1 faces comes from the opposite side of the box, where the background azimuthal velocity differs by $q\Omega L_x$, the boundary exchange must offset the wrapped azimuthal momentum and energy:

$$M_y \rightarrow M_y + \rho \Delta v, \quad E \rightarrow E + M_y \Delta v + \frac{1}{2} \rho \Delta v^2, \quad \Delta v = \pm q\Omega L_x \quad (3)$$

(+ at the inner face, − at the outer). Initial conditions must include $\mathbf{v}_0$. The two frames are the same physics on different variables; comparing them is a strong consistency test (Sec. 5). The cost of the full-velocity frame is the timestep,

$$\frac{\Delta t_{\text{FARGO}}}{\Delta t_{\text{full}}} \approx \frac{c_s + q\Omega L_x/2}{c_s + |\mathbf{v}'|_{\text{max}}}, \quad (4)$$

measured directly as $5.7\times$ for the $L_x = 2\pi$ swing-test box (Sec. 5.1) and reaching $10\text{--}30\times$ for gravito-turbulence boxes — plus the extra truncation diffusion of advecting structures through the grid at many c_s . Non-FARGO is therefore the correctness baseline, not the production mode.

2.2 Orbital advection and its refinement problem

The orbital-advection step shifts every x -column of every MeshBlock by the background displacement $\delta y(x) = q\Omega x \Delta t$, as an integer number of cells plus a conservative fractional remap,

$$q_j \rightarrow q_{j-j_{\text{off}}} - (F_{j+1-j_{\text{off}}} - F_{j-j_{\text{off}}}), \quad j_{\text{off}} = \left\lfloor \frac{\delta y}{\Delta y} \right\rfloor, \quad (5)$$

with upwind remap fluxes F of first (`dc`), second (`plm`), or third (`ppmx`) order — the same kernels as the shear-periodic boundary remap. The required y -ghost data travels in buffers of depth $n_g + \text{maxjshift}$ exchanged with the two x_2 -face neighbors.

The crucial observation for refinement: *the shift kernel is already per-block correct on a multilevel mesh*. Each block computes δy from its own cell centers and applies it in its own Δy ; blocks at different levels shift by the same physical displacement expressed in their own cell units, and after the shift the ordinary AMR ghost exchange (with prolongation/restriction at x_1/x_3 level interfaces) restores consistency to interpolation order, since a uniform-in- y translation commutes with the interpolations to that order. What is *not* level-safe is the communication pattern: the buffer exchange assumes both x_2 -face neighbors are same-level, same-size blocks, which fails at any coarse-fine interface in y . This is exactly why refinement previously worked only with orbital advection disabled.

2.3 Refinement policies

Rather than generalizing the y -communication to level interfaces, Phase 0 removes the interfaces:

Uniform-boundary-level policy (always). All MeshBlocks on the two shear-periodic x_1 boundary faces must share *one* refinement level. The reason is structural: the unshifted wrap across x_1 goes through the ordinary neighbor machinery (the tree treats `shear_periodic` exactly like `periodic`), and the shear pass then redistributes the wrapped ghost data azimuthally *within each face* —

with per-block Δy arithmetic and a level-aware target lookup (`FindTargetMB` shifts lx_2 modulo the block count at the source block’s own level). Nothing in that machinery references the root grid; its only real assumption is same-size blocks along each face and a same-level wrap between the faces. Hence the common level may exceed root: refining *both* boundary faces uniformly (full x_2/x_3 , same level) is legal, and a mixed-level boundary is a fatal error. Enforced block-by-block at tree build and restart (`Mesh::CheckShearingBoxRefinement`). Truly mixed-level shear boundaries (a refined patch on one face wrapping onto coarse blocks of the other) would require combining the y -shift with prolongation/restriction in the shear comms — deferred; no production code implements it.

Annular policy (when FARGO is on). Every refined region must span the full x_2 extent over its x_1 – x_3 footprint: refinement comes in *rings*. Within a ring every x_2 -face neighbor is same-level and same-size, so the per-level shift plus the *existing* communication code are valid with no changes. Enforced by ring-completeness counting over (level, lx_1 , lx_3); a partial ring with FARGO on is a fatal error naming the offending ring. For AMR, refine/derefine flags are synchronized ring-wise after the global flag exchange — refine the whole ring if any member is flagged, derefine only if every member is — deterministically on all ranks. Scientifically the ring shape is a good match for the target problem: GI spiral arms are azimuthally extended, and a ring at the arm’s radius captures the whole arm and every clump in it.

Buffer-depth safety. The integer shift on a refined block at radius x is $q\Omega|x|\Delta t/\Delta y_\ell$ cells, which the construction-time estimate `maxjshift` = $\lfloor \text{cfl} \cdot q\Omega \max|x| \rfloor + 1$ (now including the previously missing $q\Omega$ factor, doubled on multilevel meshes) may underestimate in unusual configurations. The unpack step therefore re-verifies the actual worst-case shift against the communicated buffer depth every step and aborts with a diagnostic rather than silently corrupting data.

3 What was built

Changes live in commits `e7ef37f2` (infrastructure) and `f1468da3` (bin outputs + docs); Phase 0a is ported from the `athenak` branch `test/shwave-smr`, with provenance kept in the input-file comments. Phase 0c adds commits `f0acd383` (derefinement-sort fix), `b5388c02` (AMR rebuild machinery, graded cap, moving-ring test), and `097a97ee` (vortical-shwave AMR test).

File (under src/)	Change
hydro/hydro.cpp, mhd/mhd.cpp	orbital_advection toggle; MHD+refinement fatal
shearing_box/shearing_box.{hpp,cpp}	store the toggle; SetX1BdryMBs / ReinitAfterMeshUpdate: rebuild boundary GID lists, MPI requests, and (via virtual AllocateBuffers, grow-only) CC/FC buffers after every AMR update
shearing_box/shearing_box_srcterms.cpp	dual-frame source terms, Eqs. (1)–(2)
shearing_box/shearing_box_cc.cpp	non-FARGO BC momentum/energy offset, Eq. (3)
shearing_box/orbital_advection.cpp	maxjshift estimate: $q\Omega$ factor, $2\times$ multilevel headroom
shearing_box/orbital_advection_cc.cpp	runtime shift-vs-buffer-depth guard (fatal)
mesh/mesh.cpp	blanket prohibition $\rightarrow$ policy warning
mesh/build_tree.cpp	Mesh::CheckShearingBoxRefinement: interior- x_1 and ring completeness
mesh/mesh_refinement.{hpp,cpp}	boundary-block refinement veto + graded cap, Eq. (6); ring-wise AMR flag sync; ReinitAfterMeshUpdate calls after each update; derefinement-sort end-iterator fix (upstream bug)
pgen/tests/shwave.cpp, swing.cpp	background shear in ICs when orbital_advection=false; MovingRingRefine user AMR criterion (amr_moving_ring)
inputs/shearing_box/ (repo root)	*_smr*.athinput, epicycle*.athinput, *_amr_ring*.athinput test inputs (bin outputs)

Runtime options (<shearing_box> block): `orbital_advection = true|false` (default `true`). Everything else is policy, enforced automatically. The gravity solver selection of the companion document (<gravity> `solver = multigrid|fft`) is unaffected by any of this; Phase 1 will add `shear_periodic` boundary support to the multigrid driver behind the same input block.

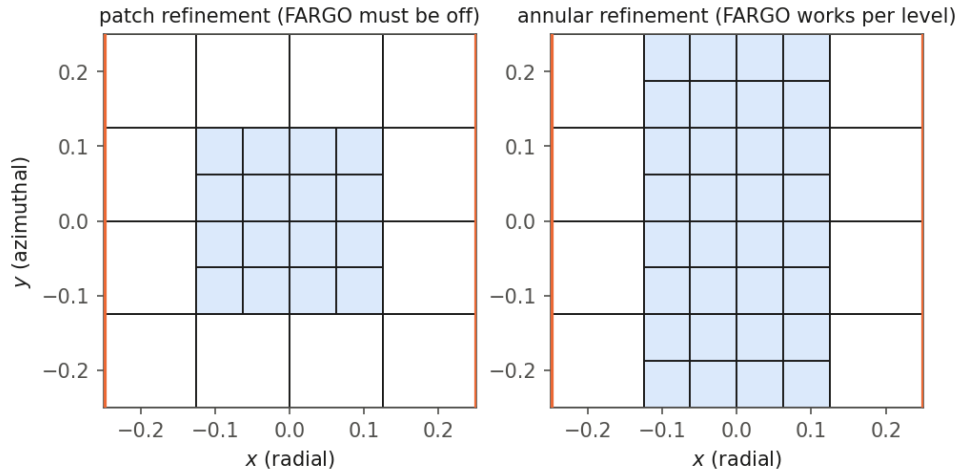


Figure 1: The two refinement layouts, drawn from the `bin` output block structure of actual runs (orange lines: shear-periodic x_1 boundaries; shaded: level-1 blocks). Left: a patch interior in both x_1 and x_2 — legal only with `orbital_advection=false`, because the coarse-fine interfaces in y break the shift communication. Right: the annular layout — the ring has no level jump in y , so FARGO runs per level with unmodified communication.

4 Building and compiling

Phase 0 needs no external dependencies beyond AthenaK’s vendored Kokkos:

```
cd athenak-multigrid
git submodule update --init kokkos      # once
mkdir build && cd build
cmake ..                                # add -D Athena_ENABLE_FFT=ON for the FFT
make -j8                                # oracle / swing+gravity cross-checks
```

Notes: SMR/AMR runs need `<mesh> nghost  $\geq 4$` (even, for prolongation); the FFT flag is only required for the tests that put gravity in the loop (it fetches kokkos-fft and needs `brew install fftw` on macOS, as detailed in the companion document). All test inputs referenced below are under `inputs/shearing_box/` and `inputs/tests/`; runs in this document were executed from `validation/run/` and `validation/run_epicycle/`.

5 Tests and results

5.1 Frame consistency: FARGO vs. full-velocity, uniform grid

```
ATH=build/src/athena
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_nofargo \
    shearing_box/orbital_advection=false
# with gravity in the loop (requires FFT build):
$ATH -i inputs/tests/swing_selfgrav.athinput job/basename=SwingNoFargo \
    shearing_box/orbital_advection=false
```

The two frames agree on the frame-independent diagnostics of the JG05 vortical shearing wave — mass exactly, x -kinetic-energy history to 3.3% (the truncation gap between two different discretizations of the same problem). With self-gravity, the non-FARGO swing test tracks the linearized shearing-sheet solution with $L_2(\delta) = 1.2 \times 10^{-2}$ (FARGO: 0.9×10^{-2}) and phase-locking of the density to the instantaneous shearing wavevector at the 10^{-12} level; it costs $5.7\times$ more cycles, exactly Eq. (4).

5.2 Refinement without FARGO (Phase 0a)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
    shearing_box/orbital_advection=false
```

An interior level-1 patch (Fig. 1, left) in the JG05 vortical-shwave problem. The run reproduces the reference run from the original `test/shwave-smr` branch **bit-for-bit** in every history column — the port changed nothing.

5.3 Guard checks (must abort)

```
# refined region snapping to a shear-boundary block: interior-x1 policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
    refined_region1/x1min=-0.24 refined_region1/x1max=-0.13
# partial-x2 ring with FARGO on: annular policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
```



```
shearing_box/orbital_advection=true
```

Both abort with explanatory fatal errors; the second names the offending ring (“ring at level=3 lx1=2 lx3=0 has 4 of 8 MeshBlocks”).

5.4 FARGO on an annular ring (Phase 0b)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
      job/basename=ring_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
      job/basename=ring_nofargo shearing_box/orbital_advection=false
```

The full- x_2 level-1 ring of Fig. 1 (right). With FARGO active on the refined mesh: mass conserved exactly (drift 0.0); x -kinetic-energy history agrees with the non-FARGO twin to 3.8% — the same truncation-level gap the two frames show on uniform grids — and with the uniform-grid FARGO run to 4.1%.

5.5 Refined x_1 boundary faces

The uniform-boundary-level rule allows refinement *at* the shear-periodic boundaries when both faces are refined uniformly. The test input refines the leftmost and rightmost root columns to level 1 over the full x_2/x_3 extent (boundary annuli), interior at root, orbital advection on:

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
      job/basename=epi_bndry problem/ipert=1
# guard (must abort): refine only ONE face -> mixed levels across the wrap
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
      refined_region2/x1min=0.0 refined_region2/x1max=0.12
```

Results: the epicycle matches the analytic solution to 5.0×10^{-6} — *identical to the last digit* with the uniform-grid and interior-ring values, so the level-1 shear-periodic wrap and the level-1 azimuthal redistribution preserve uniform fields exactly. The 20-orbit endurance run on this mesh (epicycle_smr_bndry.athinput) is *bit-identical* to the interior-ring endurance run over all 202 history outputs: for a uniform field the refined shear-boundary wrap contributes exactly nothing, the strongest available statement for this configuration. The vortical shwave conserves mass exactly and agrees with the uniform run to 4.9% in x -KE (interior ring: 4.1%) and with the interior-ring run to 0.9%; a three-panel density movie of uniform | interior ring | boundary annuli (make_bin_movie_compare.jl) shows the sheared wave crossing the refined boundary with no visible artifacts. The single-face guard aborts with a message naming the level span.

5.6 Two-level refinement (levels 1 and 2)

Both policies recurse in level, so nothing new is required for deeper hierarchies — verified with two configurations at levels 1+2 (8 root + 32 level-1 + 256 level-2 blocks each, FARGO on): nested interior rings (hydro_incompress_shwave_smr_ring2.athinput) and level-2 boundary annuli with level-1 buffer rings inboard (hydro_incompress_shwave_smr_bndry2.athinput). The epicycle matches the analytic solution to 5.0×10^{-6} on both meshes and the two runs are *bit-identical* to each other; the vortical shwave conserves mass exactly and agrees with the uniform run at truncation level (4.5% and 5.2% in x -KE; 0.45% between the two-level and one-level rings). The

three-panel density movie (uniform | nested rings | two-level boundary annuli) shows no artifacts at any level interface or at the refined shear boundary.

5.7 The endurance test: epicycles for 20 orbits

The `ipert=1` epicycle has spatially uniform $v_x(t)$, $v_y'(t)$: an exact nonlinear solution that any correct y -shift must preserve *exactly*, making it the sharpest probe of the per-level shift and its ring-boundary communication — any indexing or buffer defect appears as $O(1)$ corruption. Two long runs ($L = 20$, $128^2 \times 4$ effective resolution, ideal EOS with $c_s = \sqrt{5/3}$, $v_x(0) = 0.1 = 0.077 c_s$, 20 orbits = 20 epicyclic periods since $\kappa = \Omega$ at $q = 3/2$):

```
$ATH -i inputs/shearing_box/epicycle.athinput      # uniform 128x128x4
$ATH -i inputs/shearing_box/epicycle_smr_ring.athinput  # level-1 ring + FARGO
```

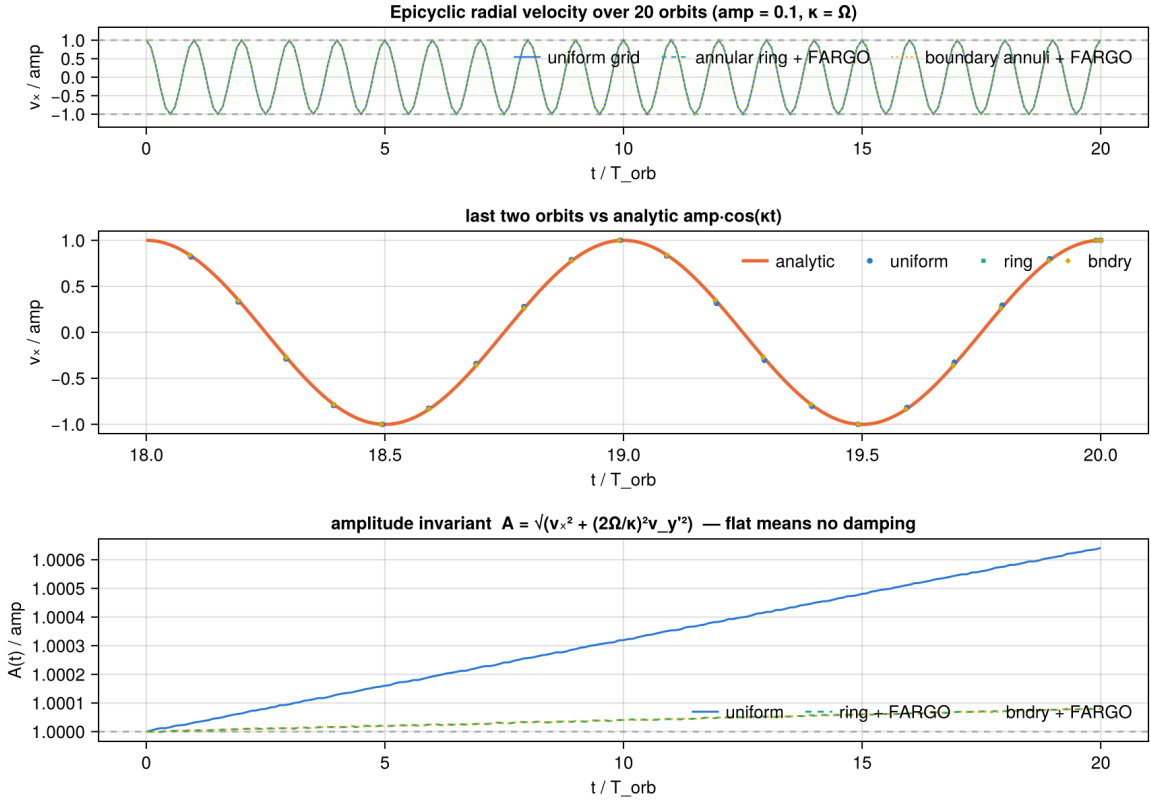


Figure 2: Volume-averaged radial velocity of the epicycle test over 20 orbits, normalized to the initial amplitude. Top: both runs oscillate between ± 1 with no visible decay. Middle: the final two orbits against the analytic $v_x = A \cos \kappa t$. Bottom: the epicyclic amplitude invariant $A = [v_x^2 + (2\Omega/\kappa)^2 v_y'^2]^{1/2}$, which isolates numerical damping/growth from phase error.

run	v_x range / amp	amplitude drift, 20 orbits	max $ v_x - \text{analytic} $ / amp
uniform 128^2	$[-1.0006, +1.0005]$	+0.064%	2.4%
annular ring + FARGO	$[-1.0000, +1.0001]$	+0.008%	0.63%
boundary annuli + FARGO	<i>bit-identical to the annular ring, all 202 outputs</i>		

Both runs hold the ± 1 envelope; the amplitude invariant is flat to a few $\times 10^{-4}$ (uniform) and 10^{-4} (ring) — a slow secular *growth* characteristic of RK2 on an oscillator, not shearing-box or refinement physics. The 3.3% peak difference between the runs is pure phase drift: the ring run takes smaller steps (finer cells) and accumulates *less* RK2 phase error, which is why the refined-mesh FARGO run is the one closer to the analytic solution. A defect in the ring communication would instead show as amplitude noise — none is present at the 10^{-5} level.

5.8 AMR in the shearing box (Phase 0c)

AMR rennumbers MeshBlock GIDs and re-assigns blocks to ranks on every mesh update, so the shear-boundary communication state built at construction — the `x1bndry_mbgid` GID lists, the MPI request arrays sized to them, and the send/recv buffers — goes stale. `ShearingBox::ReinitAfterMeshUpdate` now rebuilds all of it from the current MeshBlockPack, called from `AdaptiveMeshRefinement` after the post-update policy check and before any boundary exchange on the new mesh; the startup fatal guard is gone. Orbital advection needs no rebuild (its buffers are neighbor-indexed and sized to `max_nmb_per_rank` from the start).

Two defects surfaced by the multi-level tests, both now fixed, are worth recording. First, vetoing refinement of the boundary blocks themselves is not enough: a criterion refining blocks *near* the boundary can force refinement *of* the boundary columns through 2:1 balancing. The veto is therefore extended to a *graded cap*: a block at level $s \geq 1$ above the (uniform) boundary level may be refined only if there is room for the intermediate buffer columns between it and the boundary, which in integer logical coordinates at the block’s own level reads

$$2^{s+1} - 1 \leq l_{x1} \leq n_{mb,x1} - 2^{s+1}. \quad (6)$$

Any post-update policy violation is therefore a genuine bug rather than a criterion asking for too much. Second, an upstream AthenaK bug: the derefinement list was sorted with end iterator `&c1lderef[ctnd-1]`, which *excludes the last element*. When a mixed-level derefinement left a fine-level parent in the unsorted last slot, coarser parents merging before it were vetoed by the tree’s finer-neighbor check order-dependently — some parents of an annulus merged and some did not, splitting what must be a ring-complete derefinement (the policy check caught it as a partial ring, e.g. 4 of 8 members). Outside the shearing box the bug merely delays some derefinements by one AMR interval, which is why it went unnoticed upstream.

5.8.1 Dynamic ring creation and destruction: the moving-ring criterion

The dynamic exercise uses a deterministic user AMR criterion (`MovingRingRefine` in the `shwave` pgen): refine every MeshBlock whose x_1 extent overlaps $[x_c(t) - w, x_c(t) + w]$ with $x_c(t) = x_{c0} + A \sin(ft)$, derefine all others. The criterion depends only on x_1 , so it naturally flags complete rings; the ring-sync logic and the vetoes above do the rest. With A chosen so the interval sweeps past the (vetoed) boundary columns, *all* refined rings are destroyed near each turning point and re-created on the way back — one full create/destroy cycle per $2\pi/f$.

```
$ATH -i inputs/shearing_box/epicycle_amr_ring.athinput      # 20 orbits, 1 level
$ATH -i inputs/shearing_box/epicycle_amr_ring2.athinput     # 2 orbits, 2 levels
$ATH -i inputs/shearing_box/epicycle_amr_ring_nofargo.athinput # 2 orbits, non-FARGO
```

The 20-orbit epicycle run ($x_c = 7 \sin t$, $w = 1.9$, one sweep per orbit) creates 2296 and destroys 2240 MeshBlocks (Fig. 3, left); the counts and all history columns are bit-reproducible across reruns. Mass drift is exactly zero. The epicyclic amplitude invariant of §5.7 drifts by +0.015% over 20 orbits,

between the static ring (+0.008%) and the uniform grid (+0.064%), and the time-matched x -KE history differs from the *static*-ring run by at most 2.6×10^{-3} of the peak — an order of magnitude closer than either run is to the uniform grid (2.4×10^{-2}): the dynamic regridding adds essentially nothing beyond the static refined-boundary truncation already characterized in §5.7. The two-level variant (nested moving rings, 16–296 blocks, graded per Eq. (6)) drifts by 0 to six digits over 2 orbits; the non-FARGO variant exercises the momentum-offset boundary path with exact mass conservation. Two-panel movies with per-frame mesh outlines (`dens_compare_epi_unif_vs_epi_amr{,2}.mp4`) show the rings sweeping, collapsing, and re-forming with no visible artifacts.

5.8.2 Wave structure through a moving refinement boundary

The epicycle is spatially uniform, so it never exercises prolongation/restriction on real structure. The vortical-shwave AMR test (`hydro_incompress_shwave_amr_ring.athinput`) runs the JG05 problem of §5.1 under a moving ring ($x_c = 0.2 \sin t$, $w = 0.05$), so wave structure is prolonged, restricted, and shear-remapped across every regrid event, including full ring destruction near the turning points at 64^2 :

resolution	peak x -KE err. (static ring)	peak x -KE err. (moving ring)	AMR vs. static ring	mass drift
64^2	−4.05%	−8.55%	4.7×10^{-2}	0.0
128^2	−0.29%	−0.27%	1.8×10^{-4}	0.0

(errors vs. the same-resolution uniform run; curve maxima relative to the peak x -KE; the 128^2 runs use CLI overrides `mesh/nx1=128 mesh/nx2=128`). The swing peak lands at $t = 2.70$ in every run (theory: 2.67). At 64^2 the moving ring roughly doubles the static-ring truncation — there the ring covers only ~ 1.6 wavelengths of the $|n_{wx}| = 8$ mode and full ring destruction repeatedly restricts fine data mid-swing. At 128^2 the moving-ring and static-ring solutions are indistinguishable (1.8×10^{-4} , Fig. 3, right): the regridding cost converges away *at least* as fast as the static refined-boundary error, with no floor. (The apparent order is inflated by geometry — at 128^2 the same physical criterion keeps a relatively tighter ring that never fully vanishes — so this is a convergence statement, not an order measurement.)

5.9 Summary

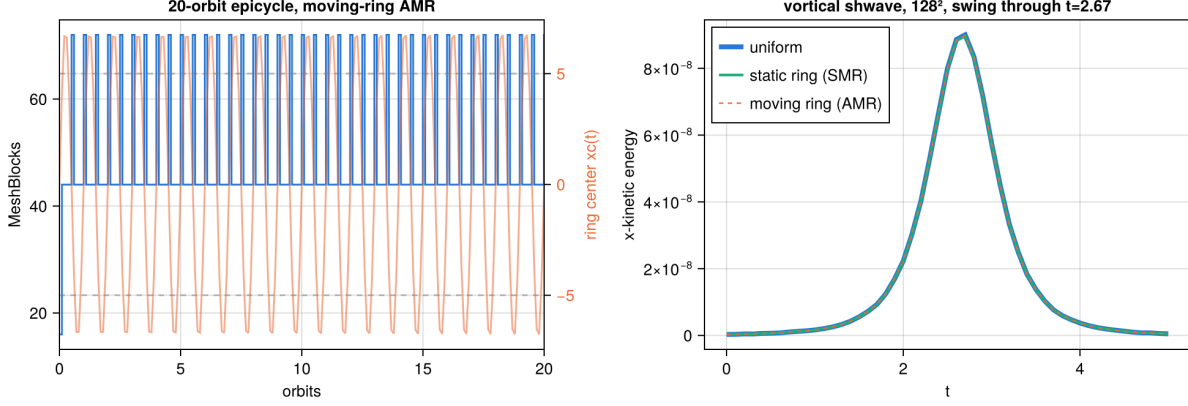


Figure 3: Phase 0c. Left: MeshBlock count (blue, left axis) in the 20-orbit moving-ring epicycle AMR run, with the prescribed ring center $x_c(t) = 7 \sin t$ (orange, right axis; dashed lines mark the vetoed boundary columns beyond $|x_1| = 5$). Rings are destroyed near every turning point and re-created on the way back, 2296 blocks created / 2240 destroyed in total. Right: x -KE history of the 128^2 vortical shwave through the swing — uniform, static SMR ring, and moving-ring AMR are visually indistinguishable (moving vs. static ring: 1.8×10^{-4} of peak).

Check	Metric	Result
SMR no-FARGO vs <code>test/shwave-smr</code> reference	all history columns	bit-identical
Uniform: FARGO vs non-FARGO	mass / x -KE	exact / 3.3%
Swing + gravity, non-FARGO vs linear theory	$L_2(\delta)$ / phase	1.2×10^{-2} / 10^{-12}
Non-FARGO cost ($L_x=2\pi$ box)	cycles vs FARGO	$5.7\times$, as predicted
Ring + FARGO: mass conservation	max drift	exact (0.0)
Shwave: ring-FARGO vs ring-non-FARGO	x -KE	3.8%
Epicycle, 20 orbits, ring + FARGO	amplitude drift / invariant	0.008% / flat to 10^{-4}
Epicycle, 20 orbits, boundary annuli	vs ring run, all columns	bit-identical (0.0)
Two-level (1+2) rings / bndry annuli	epicycle vs analytic; ring2 vs bndry2	5×10^{-6} ; bit-identical
Two-level shwave (both configs)	mass / x -KE vs uniform	exact / 4.5–5.2%
Boundary annuli + FARGO: epicycle	max rel. vs analytic	5×10^{-6} (= uniform)
Boundary annuli + FARGO: shwave	mass / x -KE vs uniform	exact / 4.9%
Mixed-level x_1 boundary	—	fatal, names the level span
Partial- x_2 ring with FARGO	—	fatal, names the ring
AMR epicycle, 20 orbits (2296 MBs created)	mass / invariant drift	exact / +0.015%
AMR epicycle vs static ring	time-matched x -KE / peak	2.6×10^{-3} (vs 2.4×10^{-2} to unif)
AMR epicycle, two-level nested rings	amp. drift, 2 orbits	0 to six digits
AMR epicycle, non-FARGO	mass	exact
AMR shwave vs static ring, $64^2 \rightarrow 128^2$	curve max rel. diff	$4.7 \times 10^{-2} \rightarrow 1.8 \times 10^{-4}$
AMR determinism	reruns, all counts/columns	bit-reproducible

6 Phase 1: shear-periodic multigrid boundary conditions

6.1 Method: an identification, not an operator change

Shear-periodicity identifies the two radial faces with a time-dependent azimuthal offset,

$$u(x, y, z) = u(x + L_x, y - q\Omega L_x t_s, z), \quad t_s = t \bmod T_{\text{shear}}, \quad T_{\text{shear}} = \frac{L_y}{q\Omega L_x}, \quad (7)$$

and nothing more: the interior 7-point Laplacian, the red-black smoother, restriction, and prolongation are all untouched. Only the x_1 ghost fill changes — the ghost column is the opposite boundary column with its content shifted by $\Delta y = \pm q\Omega t_s L_x$ (henceforth $\pm q\text{omt } L_x$; the sign convention, validated in the FFT track, is: inner ghosts shift by +, outer by -). The shift is generally a non-integer number of cells *at every multigrid level*: the same Δy is 35.52 fine cells but 4.44 cells four levels down. The integer part is index arithmetic with periodic wrap; the fractional part ϵ is applied with the same conservative remap kernels (DC/PLM/PPMX_RemapFlx) shared by orbital advection, the hydro shear boundary, and the FFT solver (order selected by the new `<gravity> mg_remap`, default `plm`).

Two structural facts make this cheap. First, the fixed point of the V-cycle (and of FAS) satisfies the *finest-level* discrete system, so the remap quality on coarse levels affects only the convergence *rate*, never the converged answer — one code path serves every level with no accuracy tax. Second, because each boundary plane is remapped as a full-period global row, the wrap supplies all guard cells and the `ngh=1` multigrid ghost width imposes no stencil restriction even for PPMX.

The shear phase is frozen once per solve as `qomt = qΩ (t mod  $T_{\text{shear}}$ )` evaluated at `pmesh->time` — the FFT solver’s convention exactly, so both solvers see identical boundary identifications at any instant (hydro’s shear BC uses $t + dt$ on the final RK stage; that $O(dt)$ offset is a pre-existing, FFT-validated convention).

6.2 The global boundary-plane algorithm

At each MeshBlock level the driver keeps two global boundary planes (full y - z extent of that level, `ngh` deep in x). One fill is three stream-ordered kernels: *gather* the interior columns adjacent to each x_1 face of the boundary blocks into the planes; *remap* each plane row by $\pm q\text{omt } L_x$ in place; *scatter* into the x_1 ghost slabs of the opposite-face blocks — over the *entire* slab $k, j \in [0, n+2\text{ngh})$, with periodic wrap in y and z supplying the slab’s edge and corner ghosts (trilinear prolongation reads the full 3^3 stencil, so corners matter). The fill is invoked at the top of `MultigridDriver::PhysicalBoundary`, which already runs after every boundary exchange in all three MG task lists including the final finest-level exchange — so the one ghost layer of Φ handed to the `SelfGravity` source term is also sheared, and no task-graph surgery was needed. It overwrites the unsheared periodic copy the neighbor communication just wrote (the tree wraps `shear_periodic` x_1 exactly like `periodic`); at `qomt = 0` the fill is skipped since that copy is already exact.

The root grid is a single replicated array, so `MGRootBoundary` gains a shear branch (`RootShearBoundaryX1`) that remaps the interior-row ghost columns in place; the existing x_2/x_3 passes then wrap the corner rows, and the 1-cell block level inherits correct ghosts through `SetFromRootGrid` unchanged. Mean-source subtraction — required for solvability — is re-enabled whenever every face is `periodic` or `shear_periodic` (it was silently disabled for any not-strictly-periodic mesh before, one of three defects that made shearing-box multigrid gravity *silently wrong* rather than unsupported: the ghosts were also overwritten with a $\Phi = 0$ Dirichlet wall after the periodic copy).

Multi-rank support is one collective at one seam: each rank gathers only its own boundary blocks’ columns into a zero-filled plane, and a single `MPI_Allreduce(SUM)` then reconstructs the global plane on every rank before the (rank-redundant) remap. Every plane cell has exactly one writer, so summing the zeros is exact and independent of reduction order — the multi-rank solve is *bit-identical* to the serial one. The root-grid fill needs no communication (the root grid is already replicated on every rank). Remaining scope is enforced by fatal guards (uniform grid, periodic x_2/x_3 , no `mg_bc` override, `mg_nghost=1`, device root grid).

A byproduct worth noting: MG task lists are now executed through a driver-independent runner, so `pgrav->Solve(nullptr, 1)` from a problem generator works with `solver=multigrid` — static

Poisson tests (below) run the multigrid solver through the identical code path as evolution.

6.3 Tests and results

All tests use the shared inputs of the FFT track with the single override `gravity/solver=multigrid` (the inputs now carry the MG parameters: `threshold=0.0` stagnation mode, FMG with one V-cycle per level, `mg_remap=ppmx`); the FFT solver runs the same input as the oracle. Multigrid requires cubic MeshBlocks, so the swing runs override `meshblock/nx*=16`.

Shear-periodic instants (plumbing arbiter). `fft_poisson_shear.athinput` with `solver=multigrid` at $t_0 = 0$ and at $t_0 = T_{\text{shear}} = 2/3$ (exercising the phase wrap) produces Φ **bit-identical** to the same solve with plain-periodic x_1 and no shearing box. Any plumbing or indexing error shows up here first.

Generic phase: residual, and what it does and does not prove. At $t_0 = 0.37$ ($\Delta y = 0.555 L_y$) the discrete residual $L[\Phi] - 4\pi G(\rho - \bar{\rho})$ evaluated with the solver-filled ghosts is 5.9×10^{-15} / 5.5×10^{-14} / 2.7×10^{-13} at $32^3/64^3/128^3$: the V-cycles converge to machine precision with no floor from the sheared boundary. But for multigrid this residual is a *self-consistency* check only — a converged solve with a wrong boundary identification would also drive it to zero (the FFT solver’s $\sim 10^{-3}$ values on the same metric measure its remap truncation against the *exact* identification instead). The accuracy arbiter is the next test.

Analytic shearing wave (the sign arbiter). `profile=shwave` initializes a single rolled-frame Fourier mode whose discrete potential is known in closed form; the pointwise Φ error is $O(1)$ under any sign or offset error. Both solvers converge; the multigrid errors are *smaller* than the FFT solver’s own, because MG remaps only the boundary ghosts while the FFT rolls and unrolls the entire field:

max rel. Φ error	32^3 ($\epsilon=0.76$)	64^3 ($\epsilon=0.52$)	128^3 ($\epsilon=0.04$)
multigrid (<code>mg_remap=ppmx</code>)	3.2×10^{-5}	4.9×10^{-6}	9.5×10^{-8}
FFT (<code>fft_remap=ppmx</code>)	1.1×10^{-4}	1.3×10^{-5}	1.6×10^{-6}

(ϵ = fractional part of the cell shift at that resolution; it varies with N , so these are not clean order measurements — the FFT track’s fixed- ϵ study established the remap orders.)

V-cycle convergence rate (the named open risk). Iterative mode (`full_multigrid=false`, warm start off a zero guess) at 64^3 , $t_0 = 0.37$ — a near-worst fractional shift $\epsilon = 0.52$ — against the identical solve at shear phase zero:

	contraction/cycle	cycles to floor	floor
periodic phase (<code>qomt = 0</code>)	0.116	15	1.28×10^{-11}
sheared ($\epsilon = 0.52$)	0.157	17	1.27×10^{-11}

The remapped boundary costs two extra V-cycles and reaches the *same* roundoff floor — no stall, no loss of the solution. The open risk is retired.

Dynamic oracle A/B: swing amplification. `swing_selfgrav.athinput` (the Phase-0/FFT-track configuration: $128^2 \times 16$, self-gravity at $4\pi G = 1.9$, two swing times) run twice, changing only the solver. Peak amplification -3.511418 in **both** runs (relative difference 5.7×10^{-8}); the full 401-point amplification histories differ by at most 2.0×10^{-7} (rms 8.6×10^{-8}); phase locking (sine projection) holds at 4×10^{-14} (MG) vs. 8×10^{-14} (FFT). At matched settings the two solvers are interchangeable.

Multi-rank (the Allreduce seam). With `-D Athena_ENABLE_MPI=ON`, the static tests across 1–8 ranks reproduce the serial results: residuals at machine precision, and the shwave error *identical to all printed digits* (the plane reconstruction is exact, so the multi-rank solve is bit-identical to serial); the `qomt = 0` shear-vs-periodic bit-identity also holds at 2 ranks. The 2-rank swing run matches the serial multigrid history to 10^{-11} — the MPI summation order of the history reductions, not the shear fill. (The rank-local diagnostics of the `fft_poisson` pgen were made MPI-aware in the process.)

Periodic regression. With no `<shearing_box>` block the code path is unchanged: the Jeans-wave evolution (`jeans_wave` vs. `jeans_fft`) reproduces the documented post-merge comparison — final density bit-identical between MG and FFT, velocities within 4×10^{-14} — and the static periodic Poisson test now also runs under `solver=multigrid` (residual 5.9×10^{-14}).

6.4 Consistency with the published Athena++ solver (Tomida & Stone 2023)

Before Phase 2 builds on this solver, we anchor it to the published results of the Athena++ multigrid method paper it descends from (Tomida & Stone 2023, ApJS 266, 7; `docs/Tomida_2023_ApJS_266_7.pdf`), reproducing their uniform-grid tests of §4.1 and §4.3 on our tree (notebook §10–11; inputs `tests/ts41_sin3.athinput` `tests/ts43_jeans.athinput`). Their §4.2 (binary potential on refined meshes) is the planned acceptance test for Phase 2.

§4.1: static sinusoidal waves. Triple-sine density ($A = 1$, $\rho_0 = 2$, $G = 1$) in a fully periodic unit box, with the analytic potential of their eq. (29). The `fft_poisson` pgen gains `profile=sin3` and a per-iteration study mode (`<problem> conv_niter`): in MGI mode successive `Solve()` calls warm-start from `pgrav->phi`, so N calls with `niteration=1` reproduce N V-cycles from the zero initial guess, and ϵ (vs. analytic, their eq. 6), δ (vs. the fully converged discrete solution, eq. 5, built by a bit-identical reference pass so $\delta_{20} \equiv 0$), and the defect d (eq. 7) are measured after every iteration inside one process; in FMG mode iteration 1 is one full FMG sweep. Results at $h = 1/16 \dots 1/256$ (their $1/512$ – $1/1024$ points exceed laptop memory):

- convergence factor per V-cycle 0.125 at 64^3 (rising to 0.127 at 256^3 , 0.088 at 16^3) in both δ and d , for both MGI and FMG — their reported ~ 0.125 (Yavneh 1996). The 0.116 measured in §6.3 on the modes+blob source is source-dependent variation, not a discrepancy;
- converged ϵ exactly second order (order 2.008 $\rightarrow$ 2.000), **identical to the FFT solver’s ϵ to every printed digit** at all resolutions (both solve the same discrete system);
- the first FMG sweep alone already *below* the converged discretization error ($\epsilon_{\text{sweep}}/\epsilon_{\text{conv}} = 0.39$ – 0.60), their “remarkable” finding.

§4.3: Jeans-wave dynamical test. Periodic $1 \times 0.5 \times 0.5$ box; the `jeans_wave` pgen’s oblique-wave machinery reproduces their $\mathbf{k} = 2\pi(1/L_x, 1/L_y, 1/L_z)$ exactly ($\lambda = 1/3$), and its `n_jeans` is their ν_{Jeans} ($\lambda_J = \sqrt{\pi c_s^2 / (G\rho_0)}$; stable case $\nu = 0.5$, $A = 10^{-6}$, $\Gamma = 5/3$, PLM + HLLC + `rk2` in place of their VL2). Two additions were needed: their *isothermal* pressure perturbation (`<problem> perturb=isothermal`; $\delta p = p_0 A \sin kx$), and their error metric (`ts_errors=true`): the rms of the L1 errors of the five conserved variables against the analytic linear solution. The isothermal perturbation is not an eigenmode — it excites a *static* self-gravitating mode ($\delta p = c_s^2 \nu^2 \delta \rho$ balancing gravity, fraction $f_{\text{st}} = (1 - 1/\Gamma)/(1 - \nu^2) = 8/15$ of the density perturbation) plus a standing gravito-acoustic wave at the dispersion-relation frequency — and the metric compares against this two-mode solution. (Its correctness is validated by the convergence itself: a wrong decomposition floors the error at $\mathcal{O}(10^{-7})$.) At $t = 1.5$ (the time of their Fig. 11; Fig. 10’s is unstated), $N_x = 16 \dots 128$:

rms of L1 errors	$N_x=16$	32	64	128
fully converged	3.60×10^{-7}	1.92×10^{-7}	3.98×10^{-8}	8.12×10^{-9}
one FMG sweep ($N_{\text{iter}}=0$)	3.99×10^{-5}	6.00×10^{-6}	1.37×10^{-6}	3.13×10^{-7}
FFT (exact potential)	3.60×10^{-7}	1.92×10^{-7}	3.98×10^{-8}	7.86×10^{-9}

Both of their findings reproduce: the converged runs are roughly second order with a slight deviation (order 2.27–2.29; $N_x=16$ is pre-asymptotic at ~ 5 cells per wavelength), while the single-sweep error is *precisely* second order (2.13) but 30–40 $\times$ larger — gravity-dominated, so “use of the fully converged solutions improves the accuracy.” The FFT solver (exact discrete potential every stage) bounds the converged curve to six digits.

6.5 Summary

Check	Metric	Result
Shear-periodic instants ($t_0 = 0$ and T_{shear})	Φ vs. periodic solve	bit-identical
Generic phase 32^3 – 128^3	discrete residual	6×10^{-15} – 3×10^{-13}
Analytic shwave, 128^3	max rel. Φ error (MG / FFT)	9.5×10^{-8} / 1.6×10^{-6}
V-cycle contraction, $\epsilon = 0.52$	per-cycle factor (vs. periodic)	0.157 (vs. 0.116)
Swing amplification vs. FFT	peak / trajectory / phase	5.7×10^{-8} / 2×10^{-7} / 10^{-14}
Multi-rank, 1–8 ranks	shwave digits / qomt=0 identity	identical / bit-identical
Jeans periodic regression	final density MG vs. FFT	bit-identical
T&S §4.1: CF / ϵ order / FMG sweep	vs. published	0.125 / 2.00 / sub-truncation
T&S §4.1: converged ϵ vs. FFT	all N , all digits	identical
T&S §4.3: converged / one-sweep orders	vs. published	2.3 / 2.13, sweep 30–40 $\times$ larger

Next: Phase 2 — multigrid gravity on refined shearing-box meshes; delivered for interior rings in §7.

7 Phase 2a: multigrid gravity on refined shearing-box meshes — interior rings

7.1 Scope and the incremental decomposition

Phase 2 combines the two halves validated so far — refinement in the shearing box (Phase 0) and the shear-aware multigrid solver (Phase 1) — in three steps that track *where refined blocks sit relative to the shear boundary*:

- **2a (this section): interior rings.** Under the Phase-0 interior- x_1 policy, refined blocks never touch the shear-periodic x_1 faces, so the boundary blocks — the only participants in the Phase-1 shear-plane fill — all remain at the root level. No octet ever needs a sheared ghost.
- **2b: boundary annuli.** Uniformly refined x_1 boundary faces (allowed by the Phase-0c-lite hydro policy) put octets *at* the shear faces and need a sheared octet ghost fill — the host-side analogue of §6.2.
- **2c: AMR.** The shear planes and offset tables are sized once at construction; adaptive refinement adds a rebuild-after-remesh step (the analogue of `ShearingBox::ReinitAfterMeshUpdate`).

The `MGGGravityDriver` constructor now enforces exactly this decomposition: with shear-periodic x_1 it aborts if refinement is *adaptive* (2c), or if any x_1 boundary MeshBlock sits above the root level (2b) — a check stricter than the hydro-side `CheckShearingBoxRefinement`, which also admits uniformly refined boundary faces. Static interior rings pass.

7.2 Step 0: the binary-potential octet control (no shear)

Upstream #776 ships the Athena++ binary-potential test (`pgen/tests/binary_gravity.cpp`, `tests/binary_grav` — precisely the mesh-refinement acceptance test of Tomida & Stone §4.2: two uniform spheres ($M_1=2$ at $x=6/1024$, $M_2=1$ at $x=-12/1024$, $R=6/1024$, 10^3 -subcell surface averaging) under four nested static refinement levels. Since none of our Phase-0/1 work had ever exercised the octet machinery (every earlier multigrid run was uniform), we run this control *first*, so that any later failure on sheared refined meshes indicts the shear coupling and not the inherited octet code.

Two caveats about the shipped input. First, its periodic boundaries are *not comparable* to the isolated two-sphere analytic solution the pgen’s error norms use (the errors come out $O(1)$ by construction); the meaningful configuration is the paper’s: outflow mesh boundaries with `gravity/mg_bc=multipole`, `mporder=4` (hexadecapole), now present as overridable defaults in the input. Second, the pgen’s printed “Potential L2” is $\sqrt{\langle |\delta\Phi/\Phi| \rangle}$ (Athena++ convention), and the `<refined_region>` blocks are *absolute physical* volumes: doubling the root resolution leaves every region at its absolute size, so between the 64^3 - and 128^3 -root runs only the innermost box gains resolution and the global mean is frozen *by construction* — region-wise norms are the meaningful metric.

With 40 V-cycles (FMG), roots $32^3/64^3/128^3 + 4$ levels (232/232/288 blocks; at 64^3 the level-1 region force-refines all root blocks, so the far field lands at $h=1/128$ in both larger runs):

rms rel. Φ error per region	R_0 (far)	R_1	R_2	R_3	R_4 (inner)
32^3 root	1.26×10^{-4}	1.58×10^{-4}	2.56×10^{-4}	3.43×10^{-4}	9.57×10^{-4}
64^3 root	3.19×10^{-5}	4.23×10^{-5}	6.55×10^{-5}	8.68×10^{-5}	2.44×10^{-4}
ratio	3.94	3.73	3.91	3.95	3.92

Second order in every region (the h of each region halves exactly between these two runs). From 64^3 to 128^3 only R_4 refines ($h = 1/1024 \rightarrow 1/2048$) and its error drops by 2.37 — saturated by the unimproved surrounding regions, whose inherited error it cannot undercut; the same-resolution regions agree between the two runs to four digits (a strong bit-level reproducibility check through the float32 snapshots). On the Tomida & Stone normalized-rms metric (their eq. 31) the 64^3 run gives 7.5×10^{-5} against their Fig. 7(d) value of $\sim 1.3 \times 10^{-4}$ at $h=1/64$ — consistent, ours slightly lower because of the force-refined far field. The late-stage convergence factor is ~ 0.5 per V-cycle, their reported refined-mesh value (§6.3 measured 0.12–0.16 on uniform grids: refinement, not shear, is what costs convergence rate). The defect reaches its 10^{-12} relative floor. The octet machinery works as published.

7.3 What 2a required — and what it did not

The entire code change is the constructor guard described above. The Phase-1 shear-plane fill needed *nothing*:

- `FillShearGhostPack` gathers and scatters only blocks whose own `mb_bcs` carry `shear_periodic` on an x_1 face. Interior fine blocks have internal (`BoundaryFlag::block`) faces there, so they are skipped automatically — their (root-unit, hence meaningless) entries in the per-block offset table are never read. No level test was needed.
- The plane geometry (`nmbx2_` × block cells per MG level) is expressed in root-grid units, and under the 2a guard every participating block *is* a root-level block, so the planes tile exactly as on the uniform grid.
- Octet x_1 neighbor tables already wrap shear-periodic like periodic (Phase-1 groundwork); under the 2a guard no octet abuts the x_1 faces, so the unsheared wrap is never consulted. (It becomes live — and wrong — only in Phase 2b, which is why boundary-level refinement stays fatal.)

SMR additionally requires `nghost=4` (AthenaK wants even ghosts with refinement) and, as always for this solver, cubic MeshBlocks; the test inputs use 16^3 blocks, giving four x_1 block columns of which the middle two are refined.

7.4 Static tests: bit-identity, the analytic arbiter on SMR, MPI

Input `tests/mg_poisson_shear_smr.athinput`: the Phase-1 shear test (§6.3) with a level-1 ring over $x_1 \in [-1/4, 1/4]$ (full x_2/x_3).

Shear-periodic instants. At $t_0 = 0$ the refined-mesh solve with shear-periodic x_1 is **bit-identical** (float64 comparison of every cell of every block) to the same mesh with plain-periodic x_1 , for both the multi-mode and shwave sources. (Compare the bin *data*: the file header echoes the input, so whole-file comparison always differs.)

Analytic shearing wave. The reference potential Φ_{ref} is the *exact solution of the discrete system on the uniform root grid* — not the continuum potential. The shwave profile initializes a single rolled-frame Fourier mode,

$$\rho = \rho_0 + A \cos(k_x x + k_y y + k_z z), \quad k_x = \frac{2\pi}{L_x} \left(n_1 + \text{qomt} \frac{L_x}{L_y} n_2 \right), \quad k_y = \frac{2\pi}{L_y} n_2, \quad k_z = \frac{2\pi}{L_z} n_3, \quad (8)$$

a plane wave tilted by the current shear phase and therefore exactly shear-periodic across the x_1 wrap. Acting on a cosine, the 7-point Laplacian has the exact (no-expansion) eigenvalue

$$\lambda_h = \frac{2 \cos(k_x \Delta x) - 2}{\Delta x^2} + \frac{2 \cos(k_y \Delta y) - 2}{\Delta y^2} + \frac{2 \cos(k_z \Delta z) - 2}{\Delta z^2}, \quad (9)$$

so $L_h \Phi = 4\pi G(\rho - \bar{\rho})$ is solved exactly by $\Phi_{\text{ref}} = [4\pi G A / \lambda_h] \cos(k_x x + k_y y + k_z z)$ at cell centers, with the Δ 's at the *root* spacing and both fields mean-subtracted (the periodic gauge). This is the closed form behind the pgen's `SHWAVE ERROR` line, inherited from the FFT track — the rolled k -space kernel *is* $1/\lambda_h$, which is why FFT solves it to machine precision. The choice fixes the attribution: on a *uniform* grid Φ_{ref} is exact, so a correct solver deviates only through its ghost-fill

remap (Fig. 4, third panel: 5×10^{-6} , and the roundoff floor at $\text{qomt} = 0$); against the *continuum* $-1/k^2$ instead, both regions would show their own $O(h^2)$ discretization error ($\sim (k\Delta x)^2/12$ per direction, 8×10^{-4} at the root spacing) and the refinement structure would blur. On the refined mesh the composite operator is *not* the root-grid operator: the fine region’s eigenvalue is closer to the continuum, differing per direction by $(kh)^2/12$ terms that sum to 6.0×10^{-4} relative at 64^3 — and the measured fine-region plateau is 6.7×10^{-4} : the fine grid deviating from Φ_{ref} by *being more accurate than it*. The pointwise maximum sits in the first fine cell at the refinement interface ($x = \pm 0.246$ at 64^3), symmetric about the box center, decaying harmonically into the coarse region, with *no* feature at the shear faces $x = \pm 1/2$; at the generic phase it is within 3% of the $\text{qomt}=0$ value at 64^3 and 0.6% at 128^3 — the interface, not the shear, owns the error:

	qomt = 0		generic phase $t_0 = 0.37$	
	max rel.	L_2 rel.	max rel.	L_2 rel.
32^3	1.18×10^{-2}	5.05×10^{-3}	1.44×10^{-2}	8.14×10^{-3}
64^3	5.51×10^{-3}	1.28×10^{-3}	5.66×10^{-3}	2.06×10^{-3}
128^3	2.66×10^{-3}	3.25×10^{-4}	2.68×10^{-3}	5.18×10^{-4}
L_2 ratios		3.94, 3.94		3.96, 3.97

L_2 is cleanly second order at both phases — the same interface behavior Tomida & Stone report (their rms metric) — while the interface *peak* converges at first order, the known local property of the mass-conserving level-boundary stencil. The shear remap contributes only the small, second-order-converging difference between the two L_2 columns. Figure 4 shows all of this at once: the error concentrates in the fine region (the reference uses the root-grid eigenvalue) and peaks on the interfaces, while the shear faces are featureless — and the same metric on the uniform Phase-1 run is three decades smaller, so the entire visible structure is refinement, not shear. One warning: the `profile=modes` residual norms print $O(1)$ on any refined mesh because the pgen evaluates a naive per-block Laplacian, which is *not* the composite multigrid operator at level interfaces; on SMR the shwave comparison is the valid metric.

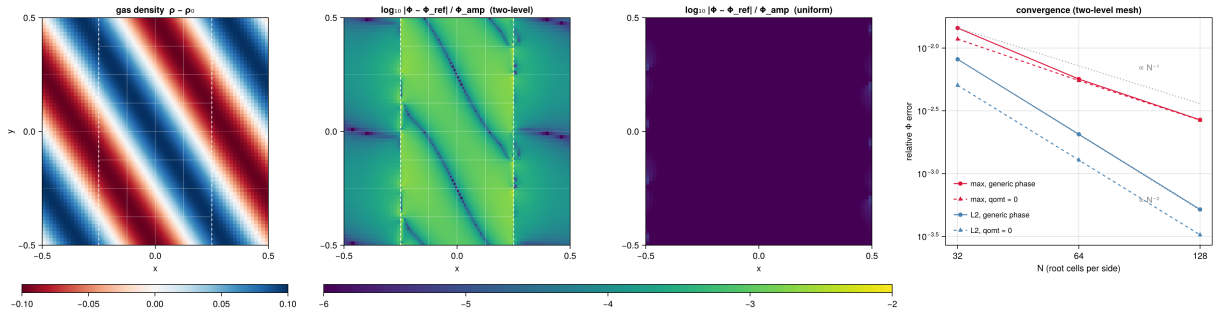


Figure 4: The static shwave test at the generic shear phase (64^3 root, level-1 ring over $|x| < 1/4$, block outlines in white). First panel: the sheared density mode. Second: $\log_{10}$ relative potential error against the root-grid closed form (eq. 9) on the two-level mesh — the error lives in the *fine* region (root–fine eigenvalue mismatch, the 6×10^{-4} plateau of §7.4), peaks in the first fine-cell layer on the refinement interfaces (dashed; bright cells modulated by the mode phase, pinched at its zero-crossing curves), and shows no feature at the shear-periodic faces $x = \pm 1/2$. Third: the *same* metric on the uniform Phase-1 run — pure remap truncation at 5×10^{-6} , three decades below the interface and eigenvalue structure. Fourth: error vs. resolution at both shear phases — L_2 second order, interface peak first order, the two phases nearly coincident.

Contraction and MPI. Static FMG solves contract at 0.18/cycle on this two-level mesh (identically for periodic and sheared boundaries; uniform grids gave 0.12–0.16). Runs at 1/2/4 ranks print identical doubles and their float32 snapshots agree to one ulp (5.8×10^{-11} , $\text{np}=2 \equiv \text{np}=4$).

7.5 Dynamic test: swing amplification on a ring, with control

Input `tests/swing_selfgrav_smr.athinput`: the §6.3 swing configuration ($128^2 \times 16$, $4\pi G = 1.9$, two swing times) with a level-1 ring over the middle half in x_1 ; the uniform twin is the single override `mesh_refinement/refinement=none`, the oracle twin additionally `gravity/solver=fft`.

	peak $d_{\cos}/A$	peak time	max $ d_{\sin} /A$
uniform, FFT	−3.511418	7.804	2.8×10^{-13}
uniform, multigrid	−3.511418	7.804	0.7×10^{-13}
ring SMR, multigrid	−3.583481	7.804	1.3×10^{-13}

The uniform twins reproduce the Phase-1 A/B exactly (the 16^3 -block relayout and `nghost=4` change nothing). The SMR peak differs from the uniform one by 2.05% — *toward* the converged answer (linear theory 3.605; the 256^2 FFT-track run gave 3.62), as refining half the domain should move it, at machine-level phase lock. The decisive attribution is the **gravity-off control** (`hydro_srcterms/self_gravity=false`, both meshes): the hydro-only SMR-vs-uniform deviation is *larger* than the self-gravitating one — peak shift +2.78% vs. +2.05%, normalized trajectory deviation 7.7×10^{-2} vs. 5.6×10^{-2} — so multigrid gravity on the refined shearing mesh contributes *no* deviation of its own beyond hydro truncation. Figure 5 shows the three amplification histories (indistinguishable at plot scale) with the pairwise differences: the SMR-vs-uniform gap rides at 10^{-2} – 10^{-1} while the MG-vs-FFT solver seam sits five decades below it; Figure 6 compares the final density fields.

7.6 The stopping rule on refined meshes (practical note)

The first full-length SMR swing run stopped at $t = 5.5$: `threshold=0.0` (stagnation-exit mode) reached the driver’s 40-V-cycle cap — *while still converging healthily* (defect 2.9×10^{-12}) — and the cap gracefully ends the run via `nlim`. The mechanism: near the swing peak the solution is dominated by the wound-up $k_x \simeq 0$, $k_y = 1$ mode, whose coarse-grid correction must pass through the sheared x_1 wrap where the y -remap acts on 8-cell rows; the warm-started contraction degrades to ~ 0.55 /cycle (static solves on the same mesh: 0.19), so the defect ratio never exceeds the 0.9 stagnation trigger within 40 cycles. This is a stopping-rule interaction, not a solver failure — and it is exactly why Tomida & Stone run production refined-mesh gravity as FMG with a *fixed* iteration count. The SMR inputs therefore default to `threshold=-1`, `niteration=10`; on the uniform twin, fixed-10 and stagnation mode agree to 10^{-11} , and the full-length SMR run completes with no MG warnings.

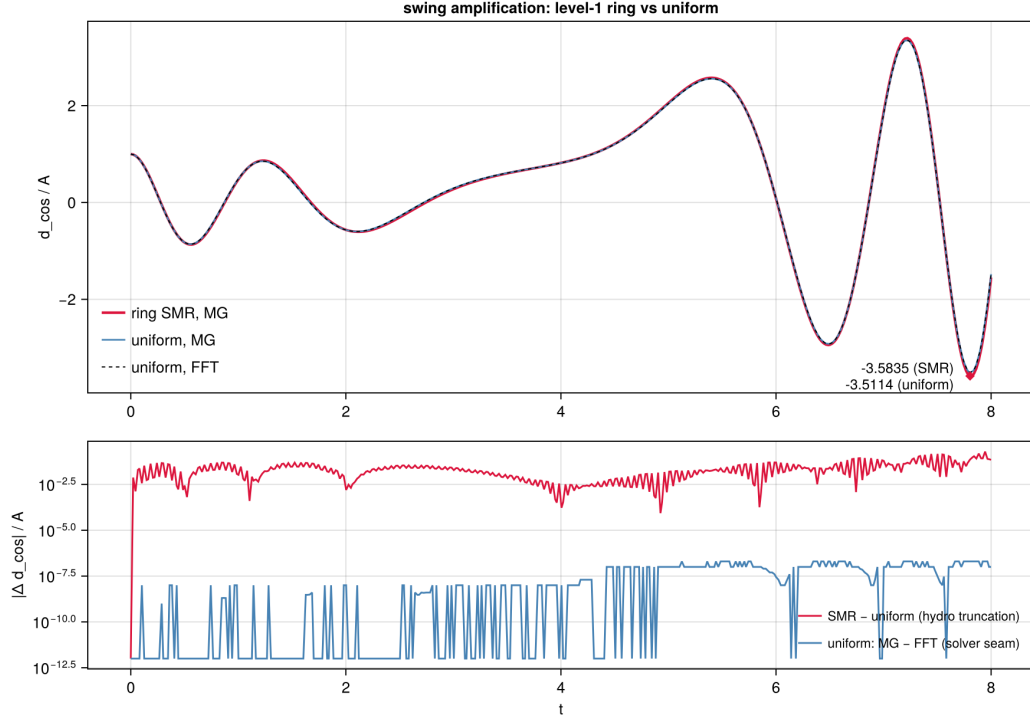


Figure 5: Swing amplification histories $d_{\cos}/A$ for the three runs (top; the curves overlap to the line width, peaks annotated) and the pairwise trajectory differences (bottom, log scale): SMR-vs-uniform is the hydro-truncation resolution blend; uniform MG-vs-FFT is the solver seam, floored by the history-file print precision.

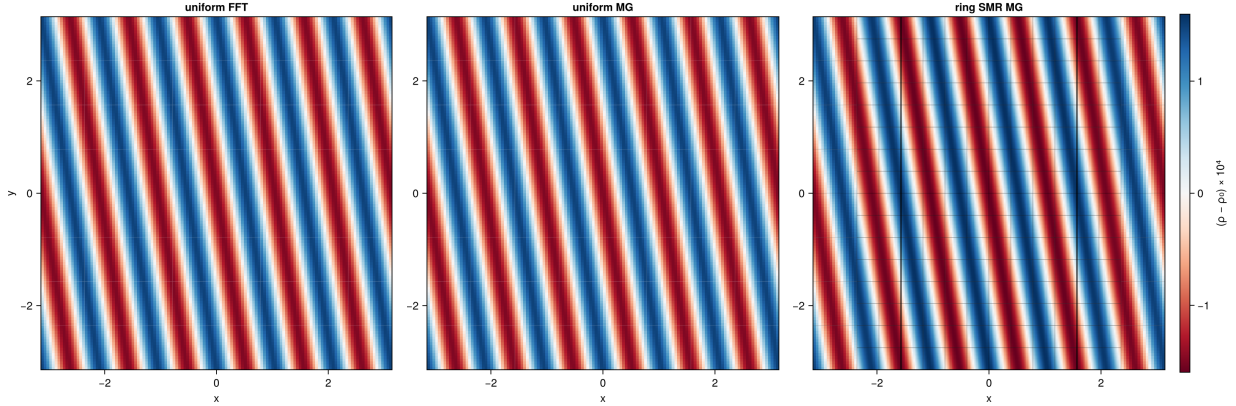


Figure 6: Final ($t = 8 = 2 t_{\text{swing}}$) density deviation $(\rho - \rho_0) \times 10^4$ at the z mid-plane for the three swing runs: uniform FFT, uniform multigrid, and the level-1 interior-ring SMR run (block outlines drawn; the heavier lines mark the refinement-level boundary). The two uniform panels are indistinguishable ($\leq 2 \times 10^{-7}$ everywhere); the SMR panel carries the same trailing wave with the amplitude difference of §7.5.

7.7 Summary

Check	Metric	Result
Binary potential (octet control)	region-wise order $32^3 \rightarrow 64^3$	3.7–3.95
Binary potential vs. T&S Fig. 7(d)	ϵ_{rms} at $h=1/64$ root	7.5×10^{-5} vs. $\sim 1.3 \times 10^{-4}$
SMR shear instants ($t_0 = 0$)	Φ vs. periodic twin	bit-identical
SMR analytic shwave	L_2 order (qomt=0 / 0.37)	3.94 / 3.96–3.97
SMR contraction (static)	per-cycle, periodic = sheared	0.18
Multi-rank 1/2/4	printed digits / f32 snapshots	identical / 1 ulp
Swing: uniform twins	peak, vs. Phase-1	–3.511418, exact
Swing: ring SMR	peak shift, direction	+2.05%, toward theory
Swing: gravity-off control	hydro-only shift (must dominate)	+2.78%
Swing: phase lock	$\max d_{\text{sin}} /A$, all runs	$\sim 10^{-13}$

Next: Phase 2b — sheared octet ghosts, so refined blocks may sit *on* the shear boundary (boundary annuli); delivered in §8.

8 Phase 2b: sheared octet ghosts — refinement on the shear boundary

8.1 Where the shear enters the octet machinery

With boundary annuli — both x_1 faces uniformly refined over the full x_2/x_3 extent, the configuration Phase 0c-lite validated for hydro — the multigrid hierarchy places *octets* (per-cell 2^3 refinement units bridging the block levels down to the root grid) directly on the shear-periodic faces. Phase 1 had already wrapped the octet x_1 neighbor tables like periodic and exempted the shear faces from the physical-boundary reflection, so octet x_1 ghosts received *unsheared* periodic values — correct only at qomt = 0, and the reason 2b was fatal-guarded. Every consumer of those ghosts (octet smoothing, the FAS right-hand side, restriction, FMG and V-cycle prolongation, and the octet→block transfer `SetFromRootGrid`) funnels through exactly two chokepoints, `SetBoundariesOctets` (per level, inside the V-cycle) and `SetOctetBoundariesBeforeTransfer` (before the root→block transfer), which makes the fix local.

Two structural facts shape the implementation. First, octets are *host-side* flat buffers, and they are *globally replicated* on every rank (the block→root transfer runs an `MPI_Allgatherv` and every rank populates every octet), so the sheared fill is plain serial host code with **no MPI seam at all**. Second, under the uniform-boundary-level policy the boundary octets tile the full y – z surface at every octet level — precisely the precondition for the Phase-1 global-plane algorithm to transplant.

8.2 The fill

`FillShearOctetGhosts(lev, folddata)` is the Phase-1 algorithm at octet granularity: gather the interior cell columns adjacent to each x_1 face from the boundary octets into a per-level (y, z) plane (level resolution $2(\text{nrbx2} \ll \text{lev})$ cells), remap each row by $\pm \text{qomt } L_x$, and scatter into the x_1 ghost slabs of the opposite-face octets — the whole slab including its y/z edge and corner ghosts (trilinear and flux-conserving prolongation read those), overwriting the unsheared values the neighbor-table exchange just wrote, with the same sign convention as `FillShearGhostPack`. The conservative remap runs through new *scalar per-face twins* of the team kernels (`DC/PLM/PPMX_RemapFlxFace` in `remap_fluxes.hpp`, host-callable, kept in documented lockstep with the validated device versions — the device path is untouched, preserving every Phase-1/2a result bit-for-bit). As in Phase 1,

octet ghost quality can only affect the convergence *rate*: the V-cycle fixed point is set by the finest block level, whose ghosts come from the (generalized) block planes.

On the block side, the only change is plane geometry: the participating blocks — the only ones whose `mb_bcs` carry `shear_periodic` — now sit at a uniform level L above root, so the planes are sized at that level (`shear_reflev_` = $L - \text{root}$); the per-block offsets were already expressed at each block’s own level. The constructor guard weakens accordingly: x_1 -face blocks must share *one* uniform level (the hydro invariant), AMR remains fatal (Phase 2c), and the deferred host-root shear fill stays guarded.

8.3 Static validation

Input `tests/mg_poisson_shear_bndry.athinput`: the §7.4 problem with the level-1 ring replaced by two annuli hugging the x_1 faces ($|x| \in [1/4, 1/2]$, full x_2/x_3); a two-level variant (level-2 annuli of width 1/16 plus the automatically graded level-1 buffers) runs off the same input via CLI overrides. The Phase-2a interior-ring results reproduce digit-exact with the new code in place (the octet fill is inert without boundary octets).

- **Shear-periodic instants.** At $t_0 = 0$ the annuli solves are **bit-identical** to their periodic twins — at one *and* two refinement levels — and their norms equal the interior-ring values to all printed digits, as they must: the two meshes differ by a half-box shift in x , under which the periodic mode’s norms are invariant.
- **Generic phase** ($t_0 = 0.37$): $L_2 = 2.057 \times 10^{-3}$, *equal to the interior-ring value to four digits* (the same half-shift symmetry, now with the remap active across the refined wrap); $\max = 5.95 \times 10^{-3}$ at the annulus’ interior interface — 5% above the ring’s peak, and still no feature *on* the shear faces themselves, now covered by fine blocks. Two-level: 6.03×10^{-3} / 4.88×10^{-3} (the larger L_2 is the root-eigenvalue reference artifact growing with off-root volume, §7.4).
- **Contraction.** Static FMG: 0.18–0.20/cycle at both one and two levels, with the *sheared* runs reaching floors slightly below their periodic twins (2.0×10^{-10} vs. 8.2×10^{-10} at one level) — the sheared octet coarse correction costs nothing.
- **Multi-rank.** $np = 2/4$ reproduce the serial digits exactly (the octet fill is replicated host work; the block planes keep their Phase-1 Allreduce).

8.4 Dynamic validation: swing on boundary annuli

Input `tests/swing_selfgrav_bndry.athinput`: the §7.5 swing configuration with the level-1 interior ring replaced by two annuli hugging the x_1 faces (2 of 8 block columns refined vs. the ring’s 4). The full-length run completes with no multigrid warnings:

	peak $d_{\cos}/A$	shift vs. uniform	$\max d_{\sin} /A$
uniform, multigrid	−3.511418	—	0.7×10^{-13}
annuli, multigrid	−3.550196	+1.10%	1.4×10^{-13}
ring (§7.5)	−3.583481	+2.05%	1.3×10^{-13}

All three peak at the same $t = 7.804$ at machine-level phase lock. The annuli shift is again *toward* the converged answer (theory 3.605), and its size scales with the refined volume — half the ring’s columns, roughly half the ring’s shift. The gravity-off control closes the attribution exactly as in

§7.5: hydro-only annuli-vs-uniform deviates *more* than the self-gravitating run (peak +1.70% vs. +1.10%; normalized trajectory deviation 6.4×10^{-2} vs. 4.3×10^{-2}) — the sheared octet machinery and level-1 shear planes add no deviation of their own beyond hydro truncation.

8.5 Summary

Check	Metric	Result
2a regression	interior-ring shwave digits	exact
Annuli shear instants, 1 and 2 levels	Φ vs. periodic twin	bit-identical
Annuli vs. ring, qomt = 0	norms (half-shift symmetry)	all digits
Annuli generic phase	L_2 vs. ring L_2	equal to 4 digits
Contraction, 1 and 2 levels	per-cycle, sheared vs. periodic	0.18–0.20, floor below
Multi-rank np = 2/4	printed digits	identical
Swing: annuli peak	shift vs. uniform, direction	+1.10%, toward theory
Swing: gravity-off control	hydro-only shift (must dominate)	+1.70%
Swing: phase lock	$\max d_{\text{sin}} /A$	1.4×10^{-13}

Next: Phase 2c — AMR in the shearing box with multigrid gravity: rebuild the block shear planes and offset tables (and re-flag the octet levels) after every mesh update, the analogue of `ShearingBox::ReinitAfterMeshUpdate`; plus the still-deferred host-root shear fill.